

Interacting ants

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi F\left(\left(X_t^i, \theta_t^i\right), \mu_t^N\right)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

- Interaction strength $\chi > 0$

- Empirical measure: $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{(X_t^i, \theta_t^i)}$, McKean-Vlasov

- Famous interacting active Brownian particle model:
Vicsek model (Phys. Rev. Lett., 1995) for flocking birds, 6000+ citations



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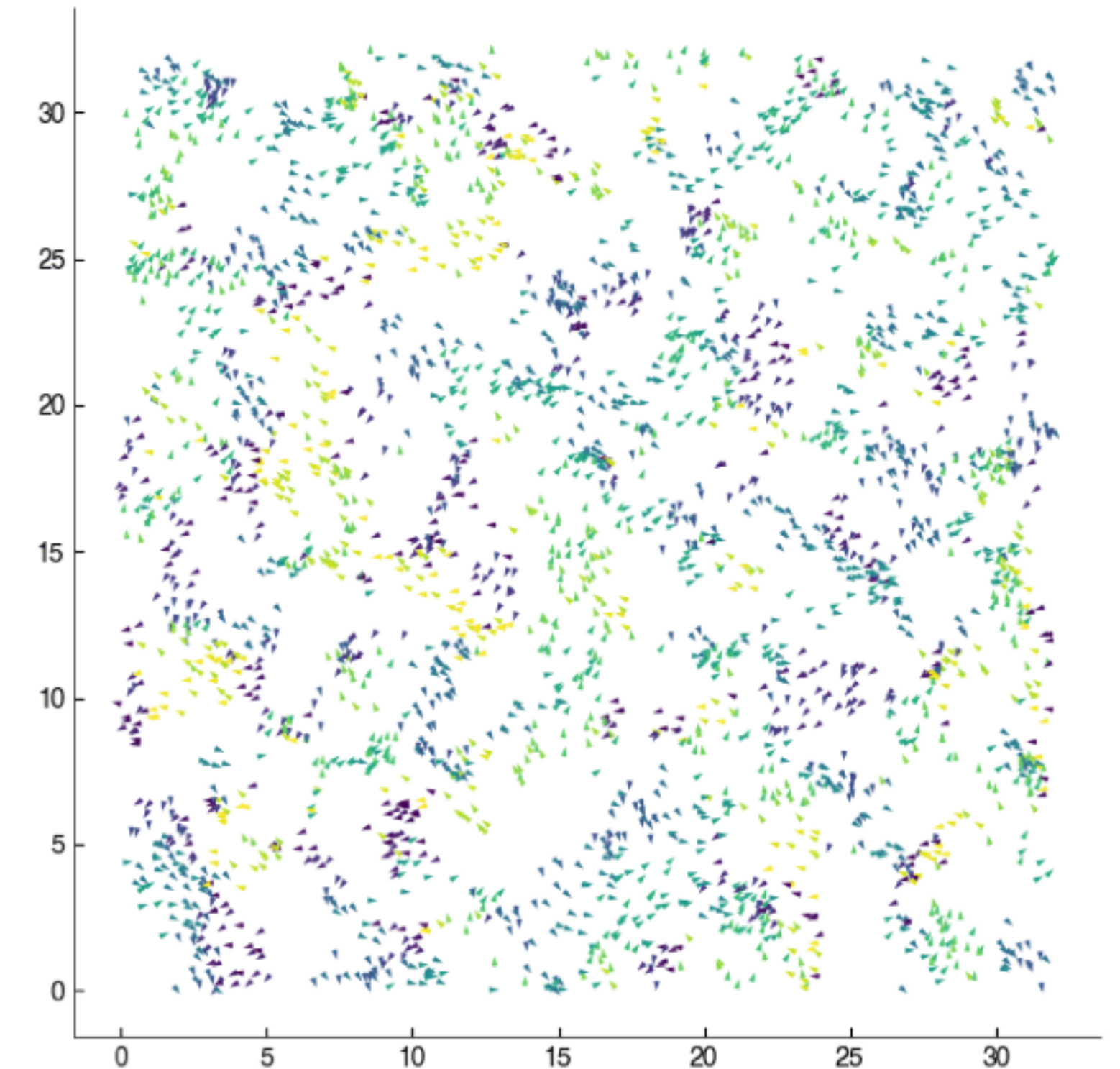
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Vicsek model simulation © F. Turci 2020